

Caribbean Community (CARICOM)

CARICOM Competition Commission

The CARICOM Competition Commission (Commission) plays a pivotal role in fostering a fair and competitive marketplace within the Caribbean Community (CARICOM) Single Market and Economy (CSME). Established under Chapter VIII of the Revised Treaty of Chaguaramas, the Commission is tasked with promoting competition, preventing anti-competitive business conduct, and ensuring consumer protection across the CSME.

Over recent years, the Commission has significantly advanced its analytical capabilities by integrating various computational tools. These tools, including STATA, SPSS, and Gretl, have enhanced the robustness and precision of the Commission's market studies and competition enforcement activities.

I. STATA: A Comprehensive Tool for Econometric Analysis

STATA is a powerful software package widely used for data analysis, data management, and graphics. In the context of the Commission's work, STATA's comprehensive suite of econometric tools enables the Commission to conduct detailed statistical analyses and model complex economic relationships.

STATA was instrumental in the Commission's research into whether mergers created efficiency gains in the banking sector in Jamaica. Using Difference-in-Difference and propensity score matching, the study showed that the effects of mergers were statistically insignificant. This finding is important as it disproves the general assumption in the region that mergers inherently create efficiencies. These insights form a crucial part of the Commission's advocacy for deeper relationships with banking sector regulators, promoting a more nuanced understanding of the impacts of mergers.

STATA was also recently utilised to investigate the factors influencing high food prices in Suriname, a country listed by the World Bank in 2023 as one of the top ten countries in the world with the highest levels of food price inflation. The Commission's research examined whether anticompetitive business conduct could be a plausible factor causing the high prices, using the software's built-in ARDL model to control for demand and supply factors. This investigation was crucial in understanding the dynamics behind Suriname's food prices and highlighting

potential areas for regulatory intervention.

II. SPSS: User-Friendly and Versatile

SPSS, known for its user-friendly interface, is another critical tool in the Commission's arsenal. It is renowned for its statistical analysis, predictive analytics, and data mining capabilities. SPSS is invaluable for conducting market surveys, analysing consumer behaviour, and segmenting markets. The software's versatility allows the Commission to perform basic and advanced statistical procedures, including descriptive statistics, hypothesis testing, and multivariate analysis. By leveraging SPSS, the Commission can glean actionable insights from data, to detect anti-competitive behaviours and assess market dynamics.

The Commission used SPSS in its research to understand the possible factors resulting in airline passengers not receiving refunds during the initial stages of the COVID-19 pandemic. This study was pivotal in identifying the challenges faced by consumers and the systemic issues within the airline industry. The results of this research and the subsequent report provided policy recommendations for the CSME region, advocating for increased consumer protection in airlines. These recommendations aimed to strengthen regulatory frameworks to protect airline customers during significant disruptions better.

The Commission also sought in this Study to identify the primary drivers of airline choice for customers when they travel. Respondents were asked to consider and score the importance of eight (8) attributes that may influence their choice of scheduled airline services. Moreover, the Commission analysed whether the mean scores were consistent when the respondents were grouped into business and leisure travelers, given the debate surrounding whether business and leisure travelers are in the same relevant market. The results showed that ticket prices and past experiences with scheduled airline services appear more important to leisure travelers than business travelers. In contrast, business travelers consider ticket flexibility more important than leisure travelers, suggesting that business and leisure travelers are in separate markets. However, the built-in SPSS independent t-test suggests that small numerical differences in mean scores for leisure and business travelers in the sample are likely by chance, and the two groups view ticket prices as equally important.

III. Gretl: Open-Source and Accessible

Gretl, an open-source software package, offers a range of econometric tools and time-series analysis capabilities. The Commission favours Gretl because of its accessibility and cost-effectiveness, especially for conducting preliminary analyses. The Commission also prefers using Gretl when seasonally adjusting time series data. This capability is essential for removing seasonal effects from data to reveal underlying trends, making Gretl an invaluable tool for time-sensitive economic analyses. The software supports statistical techniques, including linear regression, instrumental variables, and panel data analysis. The Commission utilises Gretl to perform initial data explorations, validate findings from other software, and educate staff on econometric modelling without the financial burden associated with proprietary software.

IV. Towards Machine Learning: The Future of Competition Enforcement

Recognising the limitations of traditional econometric tools in detecting complex anti-competitive practices, the Commission is poised to integrate machine learning software into its analytical framework. With its ability to uncover hidden patterns in large datasets, machine learning holds immense potential for revolutionising competition enforcement.

One of the key areas where machine learning can make a substantial impact is in detecting cartels in public procurement. Cartels often use subtle and complex strategies to manipulate bidding processes, making detection challenging with conventional methods. Machine learning algorithms can analyse vast amounts of procurement data, identifying anomalies and patterns indicative of collusive behaviour. For instance, clustering algorithms can group similar bids, while anomaly detection techniques can flag bids that deviate significantly from the norm. By adopting these advanced tools, the Commission can enhance its ability to uncover and combat cartel activities, ensuring fair competition in public procurement.

The shift towards machine learning also reflects the Commission's commitment to staying at the forefront of technological advancements in competition enforcement and assisting its limited human resources. This transition will involve not only the adoption of new software but also the upskilling of the Commission's workforce, fostering a culture of continuous learning at the Commission.

V. Conclusion

The Commission's integration of these computational tools has significantly bolstered its capacity to conduct rigorous market studies and enforce competition laws effectively.

The future adoption of machine learning software represents a promising frontier for the Commission. By leveraging the power of machine learning, the Commission can enhance its ability to detect and prevent anti-competitive practices, particularly in public procurement. This forward-thinking approach underscores the Commission's dedication to fostering a competitive, fair, and dynamic market environment within the CSME. As the Commission continues to evolve, its commitment to innovation and excellence will undoubtedly contribute to the sustainable economic development of the Caribbean region.